

Compact districting plans that better respect administrative boundaries

For California, Florida, New York, and Texas, we used a computer algorithm to find a substantially more compact (Cpt. = Polsby-Popper score) districting plans given a fixed number of times counties are divided between districts. Grey dots represent 5,000 simulated nonpartisan baseline plans.

Takeaways

- In all four states, it is possible to create more compact plans that keep more counties together than in the enacted plans.
- These plans would also create Congressional delegations that (1) are substantially more competitive and (2) tend to more closely reflect statewide voting patterns.
- However, plans with the highest levels of compactness and respect for administrative boundaries do not necessarily produce proportional representation.

How to interpret each plot

For example, in California it is possible to draw plans with 33% fewer county splits (27 → 18) that are also substantially more compact (0.21 → 0.35). Green labels indicate optimizations, while purple labels show the 2026 enacted plans. The statewide vote share shows the proportional allocation under the 2024 presidential election.

